A POWER OF THE IDEAL OF A CLOSED SUBSPACE ANNIHILATES A COHERENT SHEAF EXACTLY WHEN THE SUPPORT IS CONTAINED IN THAT SUBSPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- An affine etale cover of a quasi-compact algebraic space. Properties of Spaces, label spaces-properties-lemma-quasi-compact-affine-cover (invoked at proof line 2350). It provides the affine scheme U with a surjective etale morphism $U \to X$ over which the two conditions can be tested.
- The source of a morphism to a Noetherian space, locally of finite type, is Noetherian. Morphisms of Spaces, label spaces-morphisms-lemma-locally-finite-type-locally-noetherian (invoked at proof line 2353). It makes U a Noetherian scheme, so that the scheme case of the statement applies to it.
- The statement for schemes. Cohomology of Schemes, label coherent-lemma-power-ideal-kills-sheaf (invoked at proof line 2358). It supplies the equivalence on the Noetherian scheme U, which is pulled back to X.

Source reference. Original lemma 01Y9: sources/stacks-chow/tags.txt, source lines 2548–2548. Navigation pointer for \ref{coherent-lemma-power-ideal-kills-sheaf} used inside the counted span (line 2358); the argument that consumes it is retained in the span.

Source reference. Original lemma 04ZK: sources/stacks-chow/tags.txt, source lines 6468–6468. Navigation pointer for \ref{spaces-morphisms-lemma-locally-finite-type-locally-noetherian} used inside the counted span (line 2354); the argument that consumes it is retained in the span.

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Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Navigation pointer for \ref{spaces-properties-lemma-quasi-compact-affine-cover} used inside the counted span (line 2351); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $\mathcal{F}$ be a coherent sheaf on X. Let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals corresponding to a closed subspace $Z \subset X$. Then there is some $n \geq 0$ such that $\mathcal{I}^n\mathcal{F} = 0$ if and only if $Supp(\mathcal{F}) \subset Z$ (set theoretically).*

Proof. Choose an affine scheme U and a surjective étale morphism $U \to X$ (see Properties of Spaces, Lemma 03H6). Then U is a Noetherian scheme (by Morphisms of Spaces, Lemma 04ZK). Note that $\mathcal{I}^n\mathcal{F}|_U = 0$ if and only if $\mathcal{I}^n\mathcal{F} = 0$ and similarly for the condition on the support. Hence the result follows from the case of schemes, see Cohomology of Schemes, Lemma 01Y9. □